

# Week 1 - Lecture slides (unannotated)

Modelling, sets, and (some) algebra

HEC Montréal

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## Topics covered

- **Mathematical modeling components:** variables, parameters, constraints and assumptions.
- **Sets and algebra:** set notation, intervals, standard number sets, algebraic expressions, equations, and inequalities.
- **Proof techniques:** propositions, quantifiers, implications, necessary and sufficient conditions, five techniques of proofs: direct, contrapositive, contradiction, induction, and counterexample.

**To be completed before class:** the first two topics of mathematical modelling, sets and algebra. (Material on the course site)

**To be developed in class:** proof techniques.

**Course site:** [khaledfouda.github.io/MATH40604/](https://khaledfouda.github.io/MATH40604/). It contains all the course material needed for the English course. Since this is a new course, the material is work-in-progress and will be uploaded weekly.

**Learning objectives.** By the end of the session, you should be able to:

- state clearly the components of a mathematical model;
- translate a real-life problem into mathematical expressions and constraints;
- manipulate algebraic expressions and represent domains using sets;
- interpret quantifiers, implications, and logical conditions;
- select an appropriate proof technique and write a simple and structured proof.

# Today

**Part 0.** Introductions

**Part 1.** Review of the pre-session activities.

**Part 2.** quantifiers, implications, and Proof techniques

**Part 3.** Tutorial session

Review of pre-class material

## Mathematical modeling (Example)

We consider the market for a homogeneous good. The quantity demanded depends on the price  $p$  of the good and on consumers' income  $y$ :

$$q_D = a - bp + cy, \quad \text{where } a, b, c > 0.$$

Supply is assumed to depend on price such that  $q_S = d + ep$ , where  $d, e > 0$ . Determine the equilibrium price in the market for this good.

| Component                    | Description | Example |
|------------------------------|-------------|---------|
| Variables                    |             |         |
| Parameters                   |             |         |
| Relationships or constraints |             |         |
| Objective                    |             |         |
| Assumptions                  |             |         |

## Describing a set

**Convert each of the following to a set representation**

1.  $A = (-\infty, 5]$ . (property)
2.  $B = \{x \in \mathbb{R} \mid 1 < x \leq 4\}$ . (interval)
3. The observation space of a dataset where each observation is described by four non-negative numerical variables. (property)
4. A machine must produce a part of length 50 mm with a tolerance of 3 mm. Describe the admissible set. (interval/property)
5. A risk score  $s$  must remain within a maximum distance of 0.2 from a target equal to 0.7. Describe the admissible set. (interval/property)
6. Consider a vector of proportions  $\mathbf{x} = (x_1, \dots, x_n)$ . Each component is non-negative and the sum of the components is equal to 1. Describe the admissible set. (property)

# Sets and Summations

Consider the following weights and scores:

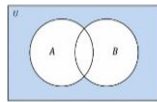
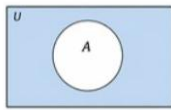
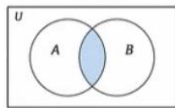
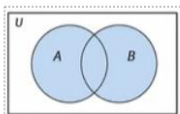
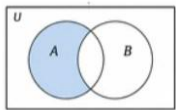
$$\mathbf{u} = (0.2, 0.5, 0.3), \quad \mathbf{z} = (70, 80, 90).$$

1. Verify that  $\mathbf{u}$  belongs to the simplex  $\Delta^3 = \{ \mathbf{w} \in \mathbb{R}_+^3 \mid \sum_{j=1}^3 w_j = 1 \}$ .
2.  $\Delta^3$  is a subset of which space (or universe)?
3. Compute the weighted score  $S = \sum_{j=1}^3 u_j z_j$ .
4. Is the vector  $\tilde{\mathbf{u}} = (0.4, 0.8, -0.2)$  admissible? Justify.
5. Explain why  $S$  lies between the smallest and the largest score when  $\mathbf{u} \in \Delta^3$ .



# Set Operations

Determine the operation that describes the blue shaded region in each diagram and present the formulas that compute the cardinality of each resulting set.



## Partition of a Universe

1. Give a mathematical definition of a partition of the universe  $U$ . (3 conditions)
2. Construct a nontrivial example of a partition of a universe of your choice and verify that it satisfies the conditions.

## Quantifiers

## Quantifiers ( $\forall$ , $\exists$ , $\exists!$ )

- $\forall$ :

Example:  $\forall x \in \mathbb{R}, x^2 \geq 0$ .

- $\exists$ :

Example:  $\exists x \in \mathbb{R}$  such that  $x^2 = 4$ .

- $\exists!$ :

Example:  $\exists! x \in \mathbb{R}$  such that  $x + 3 = 5$ .

## Note: the order of quantifiers matters

- $\forall x \in A, P(x)$ :
- $\exists x \in A$  such that  $P(x)$ :

**Which of these propositions are true?**

- $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$  such that  $y > x$ .
- $\exists y \in \mathbb{R}$  such that  $\forall x \in \mathbb{R}, y > x$ .

## Families of Sets

Let  $(A_i)_{i \in I}$  be a family of sets indexed by an index set  $I$ .

Complete the following equivalences using the quantifiers  $\exists$  and  $\forall$ :

$$x \in \bigcup_{i \in I} A_i \iff \dots$$

and

$$x \in \bigcap_{i \in I} A_i \iff \dots$$

## Implications

# Implications

$$P \implies Q$$

- $P$  implies  $Q$
- If  $P$  is true, then  $Q$  is true
- $P$  is a sufficient condition for  $Q$
- $Q$  is a necessary condition for  $P$

$$P \iff Q$$

- Both  $P \implies Q$  and  $Q \implies P$  hold
- $P$  is a necessary and sufficient condition for  $Q$  (and vice versa)
- $P$  is true if and only if (iff)  $Q$  is true



## Implications: Example

Consider the following two propositions:

- $P$ : The person resides in Montreal.
- $Q$ : The person resides in Quebec (the province).

Translate each statement into symbolic form (using  $P$ ,  $Q$ ,  $\Rightarrow$ , and  $\Leftrightarrow$ ) and determine whether it is true or false.

1. Residing in Montreal is a sufficient condition for residing in Quebec.
2. Residing in Quebec is a necessary condition for residing in Montreal.
3. Residing in Quebec is a sufficient condition for residing in Montreal.
4. Residing in Montreal is a necessary condition for residing in Quebec.
5. Residing in Montreal is a necessary and sufficient condition for residing in Quebec.

## Proof techniques

## Proof techniques

We use a mathematical proof to verify that a conclusion truly follows from the hypotheses. Proofs let trust results we did not derive ourselves.

Typically, a proof includes the following steps:

- Identify the hypotheses (what we assume to be true) and the universe (admissible set(s)) in which the object(s) are defined.
- Clearly state the conclusion to be proved.
- Construct a sequence of justified (valid) implications based on the hypotheses, definitions, and previously established results.
- Explicitly conclude that the required proposition has been proved.

Proof technique 1. direct proof

# 1. Direct Proof

We start from the hypotheses and reach the conclusion:

$$\text{hypotheses} \implies \text{logical steps} \implies \text{conclusion}.$$

**Example.** Let  $n \in \mathbb{Z}$ . Show that if  $n$  is even, then  $n^2$  is even.

**Proof.**

Proof technique 2. proof by contrapositive

## 2. Proof by Contrapositive

To prove  $P \Rightarrow Q$ , we can prove that if  $Q$  is false, then  $P$  is false.

$$(P \Rightarrow Q) \iff (\text{Not } Q \Rightarrow \text{Not } P)$$

**Example.** Let  $n \in \mathbb{Z}$ . Show that if  $n^2$  is even, then  $n$  is even.

**Proof.**

Proof technique 3. proof by contradiction



### 3. Proof by Contradiction

In contradiction, we show that the alternative is false. We want to show that  $P \implies Q$  which means that if  $P$  is true, then  $Q$  is true. In this technique, we begin the proof by assuming that  $P$  is true and  $Q$  is false, and derive a contradiction. Since  $P$  and  $\text{not}(Q)$  cannot hold together,  $P \implies Q$ .

**Example (the pigeonhole principle).** Suppose that  $n$  items are put into  $m$  containers, with  $n > m$ . Then, at least one container must contain more than one item.

**Proof.**

Proof technique 4. proof by induction

## 4. Proof by Mathematical Induction (1)

In mathematical induction, we show that our statement is true for every admissible number (or case) in the universe (universal set). This is done by first proving a simple case, then we condition on it to prove the next case. This informal metaphor found on Wikipedia helps explain the concept:

*Mathematical induction proves that we can climb as high as we like on a ladder, by proving that we can climb onto the bottom rung (the basis) and that from each rung we can climb up to the next one (the step).*

— *Concrete Mathematics*, page 3 margins.

Assume that we want to show that the statement  $P(n)$  holds in the universe  $\mathbb{N}_{\geq n_0} = \{n \in \mathbb{N} \mid n \geq n_0\}$  for some  $n_0 \in \mathbb{N}$ . The steps of a mathematical induction proof are:

- **Base case.** We first show that the first case is true. That is,  $P(n_0)$  holds.
- **Induction step.** We prove that if the statement holds for any  $k \in \mathbb{N}_{\geq n_0}$ , then it holds for  $k + 1$ . That is,  $\forall k \in \mathbb{N}_{\geq n_0}, P(k) \implies P(k + 1)$
- **Conclusion.** Given the above,  $P(n)$  holds for every  $n \in \mathbb{N}_{\geq n_0}$ .

Generally in induction, our universe will be the set of integers from  $n_0$  onwards, which automatically implies that  $k + 1$  belongs to the universe.

## 4. Proof by Mathematical Induction (2)

**Example.** Prove that  $\forall n \in \mathbb{N}_{\geq 1}$ , the statement  $P(n) : \sum_{j=1}^n j = \frac{n(n+1)}{2}$  holds, where  $\mathbb{N}_{\geq 1}$  denotes the set of positive integers.

**Proof.**

## (Dis)Proof technique 5. Counterexample

## 5. Disproof by Counterexample

Assume that we want to show that the statement  $\forall x \in E, P(x)$  does not hold, for some set  $E$ . It is sufficient to find one element  $x_0 \in E$  for which  $P(x_0)$  is false to disprove the whole statement. That is,

$$(\forall x \in E, P(x)) \text{ is false} \iff \exists x_0 \in E \text{ such that } P(x_0) \text{ is false.}$$

**Example.** Disprove the statement  $\forall x \in \mathbb{R}, x^2 > x$ .

**Proof.**

Wrap-up

# Summary

- **Mathematical Modelling.** Identify model variables, parameters, relationships, objective function, and model assumptions
- **Sets.** Describe a set (or a universe) by enumeration, by a property, or by an interval.
- **Quantifiers.**  $\forall$ ,  $\exists$ , and  $\exists!$ .
- **Functions.** How to solve equations and isolate variables using algebraic properties and the log/exponential functions.
- **Implication.**  $P \implies Q$  means that  $P$  is sufficient and  $Q$  is necessary.
- **Choosing the appropriate proof technique:**

| Technique      | Use when                                              | Assume, then derive                                      |
|----------------|-------------------------------------------------------|----------------------------------------------------------|
| Direct         | the steps from $P$ to $Q$ are straight-forward        | assume $P$ ; derive $Q$                                  |
| Contrapositive | the negations are easier to work with                 | assume not $Q$ ; derive not $P$                          |
| Contradiction  | It is easier to work with $P$ and the negation of $Q$ | assume $P$ and not $Q$ ; derive a contradiction          |
| Induction      | the claim is indexed by $n \geq n_0$                  | verify $P(n_0)$ then assume $P(k)$ and derive $P(k + 1)$ |
| Counterexample | you want to disprove $\forall x \in E, P(x)$          | assume nothing; show one example where $P(x)$ is false   |



## Before next class

- Review the material and practice the tutorial exercises.
- Prepare for the next session by reading pre-class notes of Session 2.